

## TULLN LANGENLEBARN, Austria

WMO: 110300

Lat: 48.3239N

Lon: 16.1181E

Elev: 177

StdP: 99.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.7      | -7.9 | -13.8                         | 1.2 | -9.6 | -11.1 | 1.5 | -6.7 | 14.2                  | 3.7  | 12.7 | 4.6  | 2.3                   | 250  | 0.543 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.3                   | 32.2            | 21.7 | 30.2 | 20.9 | 28.5 | 20.0 | 22.5                | 30.4 | 21.5 | 28.9 | 20.6 | 27.4 | 4.1                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.8                            | 14.9 | 26.5 | 18.9 | 14.0 | 25.2 | 18.1 | 13.3 | 24.0 | 66.8          | 30.4 | 63.3 | 29.1 | 60.1 | 27.3 | 26.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.8          | 10.3 | 9.0 | -14.4                      | 35.2 | 4.2                | 2.1 | -17.4                                              | 36.7 | -19.8      | 37.9 | -22.1      | 39.1 | -25.2      | 40.6 |    |    |
| (5)               |      |     | -14.5                      | 24.1 | 4.1                | 1.0 | -17.5                                              | 24.9 | -19.9      | 25.5 | -22.1      | 26.1 | -25.1      | 26.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.7   | 0.0   | 2.1   | 6.0   | 10.9  | 15.4  | 19.1  | 20.8  | 20.5  | 15.7  | 10.5  | 5.8   | 1.4   |  |  |
|                                                                        | DBStd       | 8.27   | 4.55  | 4.73  | 4.11  | 4.00  | 3.60  | 3.66  | 3.31  | 3.21  | 3.37  | 3.88  | 3.89  | 4.25  |  |  |
|                                                                        | HDD10.0     | 1155   | 310   | 223   | 136   | 37    | 2     | 0     | 0     | 0     | 1     | 42    | 136   | 268   |  |  |
|                                                                        | HDD18.3     | 3037   | 567   | 455   | 384   | 224   | 105   | 34    | 13    | 14    | 93    | 243   | 377   | 526   |  |  |
|                                                                        | CDD10.0     | 1417   | 1     | 1     | 10    | 64    | 169   | 273   | 336   | 324   | 171   | 58    | 8     | 1     |  |  |
|                                                                        | CDD18.3     | 257    | 0     | 0     | 0     | 1     | 14    | 57    | 91    | 81    | 13    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2540   | 0     | 0     | 0     | 21    | 151   | 581   | 903   | 769   | 112   | 3     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 843    | 0     | 0     | 0     | 2     | 24    | 184   | 338   | 275   | 20    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.7    | 4.2   | 4.2   | 4.2   | 3.9   | 3.8   | 3.4   | 3.6   | 3.2   | 3.3   | 3.3   | 3.8   | 3.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 626    | 29    | 29    | 43    | 42    | 71    | 82    | 83    | 70    | 61    | 40    | 40    | 35    |  |  |
|                                                                        | PrecMax     | 843    | 76    | 68    | 128   | 88    | 156   | 172   | 251   | 177   | 191   | 89    | 77    | 57    |  |  |
|                                                                        | PrecMin     | 424    | 2     | 1     | 7     | 0     | 19    | 32    | 10    | 17    | 14    | 3     | 1     | 9     |  |  |
|                                                                        | PrecStd     | 108    | 17    | 20    | 27    | 26    | 36    | 35    | 44    | 42    | 42    | 24    | 17    | 14    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.2  | 15.1  | 20.3  | 25.9  | 29.1  | 33.2  | 35.0  | 35.2  | 29.0  | 23.1  | 18.2  | 12.5  |  |  |
|                                                                        |             | MCWB   | 9.0   | 9.2   | 12.0  | 15.8  | 19.2  | 22.3  | 21.8  | 21.4  | 19.5  | 16.5  | 13.2  | 9.2   |  |  |
|                                                                        | 2%          | DB     | 10.6  | 12.8  | 17.8  | 23.0  | 26.8  | 30.8  | 32.3  | 32.0  | 26.2  | 20.2  | 15.2  | 10.4  |  |  |
|                                                                        |             | MCWB   | 7.5   | 8.6   | 10.8  | 14.7  | 18.4  | 21.6  | 21.7  | 21.5  | 18.5  | 15.1  | 11.9  | 8.0   |  |  |
|                                                                        | 5%          | DB     | 8.2   | 10.9  | 15.1  | 20.9  | 25.0  | 28.9  | 30.3  | 29.7  | 24.1  | 18.4  | 13.2  | 8.4   |  |  |
|                                                                        |             | MCWB   | 5.8   | 7.4   | 9.6   | 13.5  | 17.2  | 20.5  | 20.9  | 20.6  | 17.5  | 14.3  | 10.6  | 6.3   |  |  |
|                                                                        | 10%         | DB     | 6.2   | 8.9   | 12.8  | 18.8  | 22.9  | 26.9  | 28.4  | 27.6  | 22.1  | 16.6  | 11.3  | 6.8   |  |  |
|                                                                        |             | MCWB   | 4.3   | 6.1   | 8.4   | 12.3  | 16.1  | 19.4  | 20.1  | 19.7  | 16.5  | 13.4  | 9.1   | 5.2   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 9.8   | 10.1  | 12.7  | 16.6  | 20.6  | 23.6  | 23.7  | 23.4  | 20.6  | 17.1  | 13.6  | 9.7   |  |  |
|                                                                        |             | MCDB   | 12.8  | 13.9  | 19.2  | 24.3  | 27.4  | 30.9  | 32.0  | 31.6  | 27.5  | 21.2  | 17.0  | 12.0  |  |  |
|                                                                        | 2%          | WB     | 7.7   | 8.7   | 11.2  | 15.0  | 18.9  | 22.2  | 22.5  | 22.0  | 19.3  | 15.8  | 12.3  | 8.1   |  |  |
|                                                                        |             | MCDB   | 10.4  | 12.2  | 16.5  | 22.1  | 25.4  | 29.5  | 30.7  | 29.9  | 25.0  | 19.4  | 15.2  | 10.4  |  |  |
|                                                                        | 5%          | WB     | 5.9   | 7.5   | 10.1  | 13.8  | 17.7  | 21.0  | 21.5  | 21.1  | 18.2  | 14.8  | 10.7  | 6.5   |  |  |
|                                                                        |             | MCDB   | 7.9   | 10.6  | 14.2  | 20.1  | 23.6  | 27.7  | 28.9  | 28.5  | 23.1  | 17.8  | 13.1  | 8.1   |  |  |
|                                                                        | 10%         | WB     | 4.4   | 6.2   | 8.8   | 12.7  | 16.6  | 19.8  | 20.6  | 20.1  | 17.0  | 13.7  | 9.3   | 5.2   |  |  |
|                                                                        |             | MCDB   | 6.0   | 8.8   | 12.4  | 17.8  | 22.0  | 25.8  | 27.1  | 26.6  | 21.2  | 16.2  | 11.1  | 6.6   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 5.1   | 7.1   | 9.0   | 10.9  | 12.7  | 11.1  | 11.3  | 11.4  | 9.7   | 8.1   | 5.6   | 4.9   |  |  |
|                                                                        |             | MCDBR  | 7.8   | 10.6  | 14.2  | 15.1  | 14.2  | 14.5  | 15.1  | 15.4  | 13.2  | 10.9  | 8.8   | 7.5   |  |  |
|                                                                        |             | MCWBR  | 5.6   | 7.0   | 8.3   | 8.0   | 7.0   | 6.5   | 6.0   | 6.3   | 6.4   | 6.5   | 5.9   | 5.4   |  |  |
|                                                                        |             | MCDBR  | 7.7   | 9.8   | 12.9  | 14.1  | 13.1  | 13.4  | 13.7  | 13.9  | 12.5  | 9.8   | 8.4   | 7.0   |  |  |
|                                                                        | 5% WB       | MCWBR  | 5.8   | 6.6   | 7.9   | 7.6   | 6.7   | 6.3   | 6.0   | 6.1   | 6.3   | 6.2   | 5.7   | 5.3   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.318 | 0.342 | 0.383 | 0.428 | 0.427 | 0.432 | 0.439 | 0.438 | 0.409 | 0.377 | 0.349 | 0.317 |  |  |
|                                                                        | taud        |        | 2.374 | 2.330 | 2.253 | 2.160 | 2.207 | 2.232 | 2.215 | 2.236 | 2.283 | 2.344 | 2.378 | 2.403 |  |  |
|                                                                        | Ebn at Noon |        | 742   | 804   | 823   | 820   | 836   | 833   | 820   | 803   | 788   | 751   | 694   | 695   |  |  |
|                                                                        | Edh at Noon |        | 69    | 90    | 114   | 138   | 137   | 135   | 135   | 127   | 110   | 89    | 70    | 60    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.98  | 1.82  | 2.95  | 4.40  | 5.15  | 5.63  | 5.43  | 4.84  | 3.46  | 2.12  | 1.09  | 0.78  |  |  |
|                                                                        | RadStd      |        | 0.13  | 0.24  | 0.39  | 0.47  | 0.57  | 0.45  | 0.52  | 0.48  | 0.43  | 0.25  | 0.11  | 0.09  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.65 | N/A     | N/A    | N/A     | +0.53 | +0.67 | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (91 neighbors) |  | +0.54 | N/A     | N/A    | +0.80   | +0.41 | +0.34 | -121        | -152    | +75     | +38     |

Nomenclature: See separate page